Universitat Pompeu Fabra Barcelona

MTG Music Technology Group

Music Technology Lab S103

Quiz 2: Generative Models, Neural ODEs, and SCAPES

1. Generative models can create new data samples similar to the training data distribution.
   True.

2. All generative models have similar structure and they differ mostly on the data they were exposed to when training.
   False. There is a huge variety of generative model architectures.

3. Neural Ordinary Differential Equations model hidden states as continuous-time dynamics.
   True.

4. Continuous Normalizing Flows define transformations using ODEs instead of discrete layers.
   True.

5. SCAPES is a conditional generative model that learns to generate raw audio waveforms.
   False. It learns to generate audio latents (from EnCodec) rather than raw waveforms.

6. SCAPES requires manual annotations for semantic control during training.
   False. It does not require any type of manual annotations.

7. SCAPES uses CLAP embeddings as high-level semantic conditioning signals.
   True.

8. SCAPES architecture contains a large-scale transformer comparable in size to state-of-the-art diffusion models.
   False. The transformer inside of SCAPES is much smaller than most state-of-the-art models.

9. SCAPES introduces stochasticity through its flow formulation and sampling process.
   True.

10. SCAPES is designed with the assumption that some audio domains lack strong long-term structure.
    True.